Name: James Stuart

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TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 3

CLOSED BOOK, TIME: 100 MINUTES

40/40

1. Two solid spheres of identical material but different diameters R_1 and R_2 are allowed to sink freely in the same liquid. Calculate the ratio of their terminal velocities $\frac{v_2}{v_1}$, assuming $\frac{R_2}{R_1} = 4$ and that the friction factor for both is constant $f = 0.44$.

$$\boxed{\frac{v_2}{v_1} = 2}$$

Hint: Drag force of the liquid of density ρ on the sphere of radius R and velocity v is $AfK = (\pi R^2)\, f \left(\frac{1}{2}\rho v^2\right)$. The buoyant force of the liquid on the sphere is $\frac{4}{3}\pi R^3 \rho g$.

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2. Consider a Newtonian fluid with constant viscosity μ and constant density ρ flowing between two large horizontal parallel plates, a distance $2b$ apart. For $t \le 0$, flow is driven by a pressure gradient in z-direction according to the velocity profile: $v_z(x, t \le 0) = v_{z,max}\left[1 - \left(\frac{x}{b}\right)^2\right]$. At $t = 0$ the pressure gradient is turned off, after which the flow evolves in time towards stagnation.

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2.1. Simplify the equation of motion to derive a partial differential equation (PDE) for the velocity profile for $t > 0$.

$$\rho\left(\frac{\partial v_z}{\partial t} + \cancel{v_x\frac{\partial v_z}{\partial x}} + \cancel{v_y\frac{\partial v_z}{\partial y}} + \cancel{v_z\frac{\partial v_z}{\partial z}}\right) = \cancel{-\frac{\partial p}{\partial z}} + \mu\left[\frac{\partial^2 v_z}{\partial x^2} + \cancel{\frac{\partial^2 v_z}{\partial y^2}} + \cancel{\frac{\partial^2 v_z}{\partial z^2}}\right] + \cancel{\rho g_z}$$

$$\rho\frac{\partial v_z}{\partial t} = \mu\frac{\partial^2 v_z}{\partial x^2}$$

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2.2. Rewrite the PDE in terms of dimensionless variables $\phi = \frac{v_z}{v_{z,max}}$, $\xi = \frac{x}{b}$, and $\tau = \frac{\mu}{\rho b^2}t$.

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \xi^2}$$

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2.3. Write down one initial condition and two boundary conditions for the dimensionless PDE.

I.C. $\phi(\xi, \tau=0) = 1 - \xi^2$ B.C.: $\phi(\xi=-1, \tau) = 0$ $\phi(\xi=1, \tau) = 0$

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2.4. Solve the dimensionless PDE, knowing the solution may be written as an infinite sum of cosine functions $\phi(\xi, \tau) = \sum_{n=1}^{\infty} f_n(\tau)\cos\left(\frac{2n-1}{2}\pi\xi\right)$. Evaluate the integration constants using the orthogonality equation: $\int_{-1}^{1} d\xi\, \cos(n\pi\xi)\cos(m\pi\xi) = \delta_{nm}$.

Hint: $\int_{-1}^{+1}\left[(1-\xi^2)\cos\left(\frac{2n-1}{2}\pi\xi\right)\right] = -(-1)^n\frac{32}{[\pi(2n-1)]^3}$

$$\phi(\xi,\tau) = \sum_{n=1}^{\infty}\frac{-(-1)^n\, 32}{[\pi(2n-1)]^3}\left(\exp\left[-\left(\frac{2n-1}{2}\pi\right)^2\tau\right]\right)\cos\left(\frac{2n-1}{2}\pi\xi\right)$$

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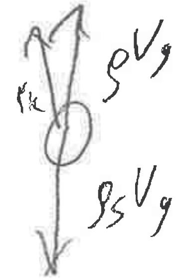
1.

at terminal velocity

$\sum F = 0$

v_t:



$\rho_s V_g = F_K + \rho V_g$

$\rho_s V_g - \rho V_g = F_K$

$V_g(\rho_s - \rho) = \pi R^2 f \frac{1}{2} \rho v_\infty^2$

$\rho V_g = \frac{4}{3}\pi R^3 \rho g$

$V_g(\rho_s - \rho) = (\pi R^2) f \left(\frac{1}{2}\rho v_\infty^2\right)$

$$\sqrt{\frac{\frac{8}{3}\pi R^3 g(\rho_s - \rho)}{\pi R^2 f \rho}} = v_\infty$$

$$\frac{v_{2\infty}}{v_{1\infty}} = \frac{\sqrt{\frac{\cancel{\frac{8}{3}}\cancel{\pi} R_2^{\cancel{3}} \cancel{g}\cancel{(\rho_s - \rho)}}{\cancel{\pi} \cancel{R_2^2} \cancel{f} \cancel{\rho}}}}{\sqrt{\frac{\cancel{\frac{8}{3}}\cancel{\pi} R_1^{\cancel{3}} \cancel{g}\cancel{(\rho_s - \rho)}}{\cancel{\pi} \cancel{R_1^2} \cancel{f} \cancel{\rho}}}} = \frac{v_{2\infty}}{v_{1\infty}} = \sqrt{\frac{R_2}{R_1}} = \sqrt{4} = \boxed{2}$$